

Zhaoxuan Tan

Ph.D. Candidate, Computer Science and Engineering, University of Notre Dame

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Summary

Final-year Ph.D. candidate (graduating May 2027) seeking full-time Research Scientist / Applied Scientist roles. My research builds **personalized LLMs**: modeling users from their data, **aligning** models with implicit user feedback through post-training, and adapting models **efficiently** per user. Five industry research internships at **Google, Amazon, and Microsoft**; 40+ papers at NeurIPS, ICLR, ACL, EMNLP, NAACL, WWW, and SIGIR, including 10 first-author papers; 3,000+ citations (h-index 24).

Education

University of Notre Dame, Notre Dame, IN 2023.08 – 2027.05 (expected)

Ph.D. in Computer Science and Engineering. Advisor: Prof. [Meng Jiang](#)

Xi'an Jiaotong University, Xi'an, China 2019.08 – 2023.07

B.E. in Computer Science and Technology. Advisor: Prof. [Minnan Luo](#)

Industry Research Experience

Microsoft, Office of Applied Research, Redmond, WA 2026.05 – present

Research Intern. Hosts: Dr. [Pei Zhou](#), Dr. [Mengting Wan](#)

- Developing group cognition memory for LLMs, extending personalization from individual users to groups so that the model can help group members collaborate.

Google, CoreML, Remote 2025.10 – 2026.04

Student Researcher. Hosts: Dr. [Yan Zhu](#), Dr. [Xiang Zhai](#)

- Built **BLUE**, an RL framework that trains an LLM profiler to write textual user profiles, using an embedding-based recommender as the reward so that profiles move toward positive items and away from negative ones.
- Added a next-item-prediction signal in text space so that profiles stay interpretable while remaining useful for retrieval. BLUE beat strong baselines on zero-shot sequential recommendation (Amazon Reviews 2023, Google Local), with clear cross-domain gains. First-author paper at **NeurIPS 2026**.

Amazon, Rufus, Palo Alto, CA 2025.06 – 2025.10

Applied Scientist Intern. Hosts: Dr. [Zixuan Zhang](#), Dr. [Haoyang Wen](#)

- Proposed **Profile-to-PEFT**, a hypernetwork trained end to end that maps a user profile directly to a full set of LoRA adapter parameters. This removes per-user training at deployment and generalizes to unseen users.
- Outperformed prompt-based personalization and per-user fine-tuning (OPPU) while using substantially less deployment compute. First-author paper at **ACL 2026**.

Microsoft, Office of Applied Research, Remote 2025.03 – 2025.06

Research Intern (part-time). Hosts: Dr. [Pei Zhou](#), Dr. [Mengting Wan](#)

- Built the Meeting Dynamic Forecasting Benchmark (MDFB) from 2,207 real-world meetings and 24,794 forward-looking queries. It tests whether LLMs can continue multi-party discussions while tracking agendas, roles, and disagreement.
- Contributed to **GLARE**, an adversarial-reward post-training method that beat SFT and SPIN in human evaluation (0.66 win rate on utility, 0.70 on human-likeness).

Amazon, Rufus, Palo Alto, CA 2024.05 – 2024.10

Applied Scientist Intern. Hosts: Dr. [Zheng Li](#), Dr. [Tianyi Liu](#)

- Proposed **PUGC**, which turns unlabeled user-generated content into scalable preference data for DPO. It needs no human- or LLM-curated preference labels.
- Achieved a 9.37% gain over standard preference-data pipelines and a state-of-the-art 35.93% length-controlled win rate on AlpacaEval 2 with Mistral-7B-Instruct. First-author paper at **ACL 2025**.

Publications

* denotes equal contribution. Full list on [Google Scholar](#).

First-Author Publications

- [1] **Bridging Textual Profiles and Latent User Embeddings for Personalization.**
Zhaoxuan Tan, Xiang Zhai, Yan Zhu, Meng Jiang, Mohamed Hammad. *NeurIPS 2026*.
- [2] **Instant Personalized Large Language Model Adaptation via Hypernetwork.**
Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, Rongzhi Zhang, Pei Chen, Fengran Mo, Zheyuan Liu, Qingkai Zeng, Qingyu Yin, Meng Jiang. *ACL 2026*.
- [3] **Aligning Large Language Models with Implicit Preferences from User-Generated Content.**
Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, Hyokun Yun, Ming Zeng, Pei Chen, Zhihan Zhang, Yifan Gao, Ruijie Wang, Priyanka Nigam, Bing Yin, Meng Jiang. *ACL 2025*.
- [4] **Can Large Language Models Understand Preferences in Personalized Recommendation?**
Zhaoxuan Tan, Zinan Zeng, Qingkai Zeng, Zhenyu Wu, Zheyuan Liu, Fengran Mo, Meng Jiang. *arXiv preprint 2025*.
- [5] **Personalized Pieces: Efficient Personalized Large Language Models through Collaborative Efforts.**
Zhaoxuan Tan, Zheyuan Liu, Meng Jiang. *EMNLP 2024*.
- [6] **Democratizing Large Language Models via Personalized Parameter-Efficient Fine-tuning.**
Zhaoxuan Tan, Qingkai Zeng, Yijun Tian, Zheyuan Liu, Bing Yin, Meng Jiang. *EMNLP 2024*.
- [7] **User Modeling in the Era of Large Language Models: Current Research and Future Directions.**
Zhaoxuan Tan, Meng Jiang. *IEEE Data Engineering Bulletin 2023*.
- [8] **BotPercent: Estimating Bot Populations in Twitter Communities.**
Zhaoxuan Tan*, Shangbin Feng*, Melanie Sclar, Herun Wan, Minnan Luo, Yejin Choi, Yulia Tsvetkov. *Findings of EMNLP 2023*.
- [9] **KRACL: Contrastive Learning with Graph Context Modeling for Sparse Knowledge Graph Completion.**
Zhaoxuan Tan, Zilong Chen, Shangbin Feng, Qingyue Zhang, Qinghua Zheng, Jundong Li, Minnan Luo. *The Web Conference (WWW) 2023*.
- [10] **Twibot-22: Towards Graph-Based Twitter Bot Detection.**
Shangbin Feng*, Zhaoxuan Tan*, Herun Wan*, Ningnan Wang*, Zilong Chen*, Binchi Zhang*, et al., Jundong Li, Minnan Luo. *NeurIPS 2022, Datasets and Benchmarks Track*.

Co-Authored Publications

- [11] **Prompt-Activation Duality: Improving Activation Steering via Attention-Level Interventions.**
Diancheng Kang, Zheyuan Liu, Ningshan Ma, Yue Huang, Zhaoxuan Tan, Meng Jiang. *NeurIPS 2026*.
- [12] **Steering Multimodal Large Language Models Decoding for Context-Aware Safety.**
Zheyuan Liu, Zhangchen Xu, Guangyao Dou, Xiangchi Yuan, Zhaoxuan Tan, Radha Poovendran, Meng Jiang. *EMNLP 2026*.
- [13] **DRIFT: Learning from Abundant User Dissatisfaction in Real-World Preference Learning.**
Yifan Wang, Bolian Li, Junlin Wu, Zhaoxuan Tan, Zheli Liu, Ruqi Zhang, Ananth Grama, Qingkai Zeng. *ICLR 2026*.
- [14] **From Personal to Collective: On the Role of Local and Global Memory in LLM Personalization.**
Zehong Wang, Junlin Wu, Zhaoxuan Tan, Bolian Li, Xianrui Zhong, Zheli Liu, Qingkai Zeng. *Findings of ACL 2026*.

- [15] **Agentic Conversational Search with Contextualized Reasoning via Reinforcement Learning.**
Fengran Mo, Yifan Gao, Sha Li, Hansi Zeng, Xin Liu, **Zhaoxuan Tan**, Xian Li, Jianshu Chen, Dakuo Wang, Meng Jiang. *Findings of ACL 2026*.
- [16] **MTMCS-Bench: Evaluating Contextual Safety of Multimodal Large Language Models in Multi-Turn Dialogues.**
Zheyuan Liu, Dongwhi Kim, Yixin Wan, Xiangchi Yuan, **Zhaoxuan Tan**, Fengran Mo, Meng Jiang. *Findings of ACL 2026*.
- [17] **Generalizable LLM Learning of Graph Synthetic Data with Post-training Alignment.**
Yizhuo Zhang*, Heng Wang*, Shangbin Feng*, **Zhaoxuan Tan**, Xinyun Liu, Yulia Tsvetkov. *Findings of ACL 2026*.
- [18] **GLARE: Generative Learning via Adversarial Reward Estimation for Social Dynamics Forecasting.**
Tenghao Huang, **Zhaoxuan Tan**, Muhao Chen, Jonathan May, Mengting Wan, Longqi Yang, Pei Zhou, Sihao Chen. *arXiv preprint 2026*.
- [19] **MoCo: A One-Stop Shop for Model Collaboration Research.**
Shangbin Feng*, Yuyang Bai*, Ziyuan Yang*, Yike Wang, **Zhaoxuan Tan**, Jiajie Yan, Zhenyu Lei, Wenxuan Ding, Weijia Shi, et al., Luke Zettlemoyer, Yejin Choi, Yulia Tsvetkov. *arXiv preprint 2026*.
- [20] **Adaptive Personalized Conversational Information Retrieval.**
Fengran Mo, Yuchen Hui, Yuxing Tian, **Zhaoxuan Tan**, Chuan Meng, Zhan Su, Kaiyu Huang, Jian-Yun Nie. *CIKM 2025*.
- [21] **Enhancing Mathematical Reasoning in LLMs by Stepwise Correction.**
Zhenyu Wu, Qingkai Zeng, Zhihan Zhang, **Zhaoxuan Tan**, Chao Shen, Meng Jiang. *ACL 2025*.
- [22] **Modality-Aware Neuron Pruning for Unlearning in Multimodal Large Language Models.**
Zheyuan Liu, Guangyao Dou, Xiangchi Yuan, Chunhui Zhang, **Zhaoxuan Tan**, Meng Jiang. *ACL 2025*.
- [23] **CodeTaxo: Enhancing Taxonomy Expansion with Limited Examples via Code Language Prompts.**
Qingkai Zeng, Yuyang Bai, **Zhaoxuan Tan**, Zhenyu Wu, Shangbin Feng, Meng Jiang. *Findings of ACL 2025*.
- [24] **IHEval: Evaluating Language Models on Following the Instruction Hierarchy.**
Zhihan Zhang, Shiyang Li, Zixuan Zhang, Xin Liu, Haoming Jiang, Xianfeng Tang, Yifan Gao, Zheng Li, Haodong Wang, **Zhaoxuan Tan**, Yichuan Li, Qingyu Yin, Bing Yin, Meng Jiang. *NAACL 2025*.
- [25] **Protecting Privacy in Multimodal Large Language Models with MLLMU-Bench.**
Zheyuan Liu, Guangyao Dou, Mengzhao Jia, **Zhaoxuan Tan**, Qingkai Zeng, Yongle Yuan, Meng Jiang. *NAACL 2025*.
- [26] **Empirical Guidelines for Deploying LLMs onto Resource-constrained Edge Devices.**
Ruiyang Qin, Dancheng Liu, Zheyu Yan, **Zhaoxuan Tan**, Zixuan Pan, Zhengge Jia, Meng Jiang, Ahmed Abbasi, Jinjun Xiong, Yiyu Shi. *ACM Transactions on Design Automation of Electronic Systems 2025*.
- [27] **KGQuiz: Evaluating the Generalization of Encoded Knowledge in Large Language Models.**
Yuyang Bai*, Shangbin Feng*, Vidhisha Balachandran, **Zhaoxuan Tan**, Shiqi Lou, Tianxing He, Yulia Tsvetkov. *The Web Conference (WWW) 2024, Oral*.
- [28] **Large Language Models Can Self-Correct with Key Condition Verification.**
Zhenyu Wu, Qingkai Zeng, Zhihan Zhang, **Zhaoxuan Tan**, Chao Shen, Meng Jiang. *EMNLP 2024*.
- [29] **Can LLM Graph Reasoning Generalize beyond Pattern Memorization?**
Yizhuo Zhang*, Heng Wang*, Shangbin Feng*, **Zhaoxuan Tan**, Xiaochuang Han, Tianxing He, Yulia Tsvetkov. *Findings of EMNLP 2024*.
- [30] **Chain-of-Layer: Iteratively Prompting Large Language Models for Taxonomy Induction from Limited Examples.**
Qingkai Zeng*, Yuyang Bai*, **Zhaoxuan Tan**, Shangbin Feng, Zhenwen Liang, Zhihan Zhang, Meng Jiang. *CIKM 2024*.

- [31] **What Does the Bot Say? Opportunities and Risks of Large Language Models in Social Media Bot Detection.**
Shangbin Feng, Herun Wan, Ningnan Wang, **Zhaoxuan Tan**, Minnan Luo, Yulia Tsvetkov. *ACL 2024*.
- [32] **DELL: Generating Reactions and Explanations for LLM-Based Misinformation Detection.**
Herun Wan*, Shangbin Feng*, **Zhaoxuan Tan**, Heng Wang, Yulia Tsvetkov, Minnan Luo. *Findings of ACL 2024*.
- [33] **Towards Safer Large Language Models through Machine Unlearning.**
Zheyuan Liu, Guangyao Dou, **Zhaoxuan Tan**, Yijun Tian, Meng Jiang. *Findings of ACL 2024*.
- [34] **Knowledge Crosswords: Geometric Knowledge Reasoning with Large Language Models.**
Wenxuan Ding*, Shangbin Feng*, Yuhan Liu, **Zhaoxuan Tan**, Vidhisha Balachandran, Tianxing He, Yulia Tsvetkov. *Findings of ACL 2024*.
- [35] **LMBot: Distilling Graph Knowledge into Language Model for Graph-less Deployment in Twitter Bot Detection.**
Zijian Cai, **Zhaoxuan Tan**, Zhenyu Lei, Zifeng Zhu, Hongrui Wang, Qinghua Zheng, Minnan Luo. *WSDM 2024*.
- [36] **Machine Unlearning in Generative AI: A Survey.**
Zheyuan Liu, Guangyao Dou, **Zhaoxuan Tan**, Yijun Tian, Meng Jiang. *arXiv preprint 2024*.
- [37] **Can Language Models Solve Graph Problems in Natural Language?**
Heng Wang, Shangbin Feng, Tianxing He, **Zhaoxuan Tan**, Xiaochuang Han, Yulia Tsvetkov. *NeurIPS 2023, Spotlight*.
- [38] **Detecting Spoilers in Movie Reviews with External Movie Knowledge and User Networks.**
Heng Wang, Wenqian Zhang, Yuyang Bai, **Zhaoxuan Tan**, Shangbin Feng, Qinghua Zheng, Minnan Luo. *EMNLP 2023*.
- [39] **KALM: Knowledge-Aware Integration of Local, Document, and Global Contexts for Long Document Understanding.**
Shangbin Feng, **Zhaoxuan Tan**, Wenqian Zhang, Zhenyu Lei, Yulia Tsvetkov. *ACL 2023*.
- [40] **BotMoE: Twitter Bot Detection with Community-Aware Mixtures of Modal-Specific Experts.**
Yuhan Liu, **Zhaoxuan Tan**, Heng Wang, Shangbin Feng, Qinghua Zheng, Minnan Luo. *SIGIR 2023*.
- [41] **HOFA: Twitter Bot Detection with Homophily-Oriented Augmentation and Frequency Adaptive Attention.**
Sen Ye, **Zhaoxuan Tan**, Zhenyu Lei, Ruijie He, Hongrui Wang, Qinghua Zheng, Minnan Luo. *arXiv preprint 2023*.
- [42] **PAR: Political Actor Representation Learning with Social Context and Expert Knowledge.**
Shangbin Feng, **Zhaoxuan Tan**, Zilong Chen, Peisheng Yu, Qinghua Zheng, Xiaojun Chang, Minnan Luo. *EMNLP 2022*.
- [43] **Heterogeneity-Aware Twitter Bot Detection with Relational Graph Transformers.**
Shangbin Feng, **Zhaoxuan Tan**, Rui Li, Minnan Luo. *AAAI 2022*.

Honors, Awards, and Research Grants

Tinker Research Grant, Thinking Machines Lab (\$5,000 in Tinker API credits)	2026
Lambda Research Grant, Lambda (\$5,000 in compute credits)	2026
OpenAI Researcher Access Program, OpenAI (\$10,000 in API credits)	2025
Top Reviewer, NeurIPS	2024
Excellent Bachelor Thesis (rank 1/172), Xi'an Jiaotong University	2023
AAAI Student Scholarship	2022
National Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM)	2021
Dean's List, Xi'an Jiaotong University	2020 – 2022

Professional Service

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- Senior Program Committee: AAAI 2027.

- **Area Chair:** ACL Rolling Review (2025–).
- **Conference Reviewer:** NeurIPS (2023–; Datasets & Benchmarks 2022), ICLR (2024–), ICML (2025–), ACL Rolling Review (2023–), COLM (2024–), KDD (2024–), AISTATS (2025–), The Web Conference (2024–2025), ICWSM (2023–), Learning on Graphs (2022–2024).
- **Journal Reviewer:** TMLR, IEEE TKDE, IEEE TNNLS, IEEE/ACM Transactions on Networking.
- **Workshop Reviewer:** ACL SRW 2025; ICLR AGI 2024; EMNLP WiNLP 2024; ACL KnowledgeNLP 2024; AAAI GCLR 2024; NeurIPS Temporal Graph Learning 2023.
- **Volunteer:** EMNLP 2022, 2023.

Mentorship and Leadership

Director, LUD Lab, Xi'an Jiaotong University: led a 17-person undergraduate research group. 2022 – 2023

Students mentored (paper → next position):

- Heng Wang: NLGraph (NeurIPS 2023 Spotlight), MVSD (EMNLP 2023) → Ph.D., UIUC
- Yuhan Liu: BotMoE (SIGIR 2023) → Ph.D., NYU
- Yuyang Bai: KGQuiz (WWW 2024 Oral) → Ph.D., Texas A&M
- Zijian Cai: LMBot (WSDM 2024) → M.S., Institute of Automation, CAS
- Sen Ye: HOFA → Ph.D., Peking University
- Thomas Mitchell: undergraduate researcher, University of Notre Dame

Teaching

Teaching Assistant, CSE 40113 Design and Analysis of Algorithms, University of Notre Dame Spring 2024

Teaching Assistant, CSE 30124 Introduction to Artificial Intelligence, University of Notre Dame Fall 2023